Group: Adios

Domain 2

SYSTEM ANALYSIS DOCUMENTATION (SAD)

UMHackathon 2026

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1. Introduction:

In today's digital economy, many SMEs and small online sellers face increasing pressure to make fast and accurate business decisions, especially in areas such as product selection, pricing, and market positioning. However, most of these decisions are still made based on **intuition, incomplete data, or fragmented information sources**. This often leads to poor choices, missed opportunities, and financial losses.

One of the key challenges is that SME data is typically **unstructured and inconsistent**. Business owners may rely on informal notes such as "this product is trending" or "supplier is unreliable," which are difficult to quantify and analyze systematically. Additionally, even when data is available, many users lack the tools or expertise to evaluate **trade-offs and opportunity costs** between different options.

Our strategy is to build **Ka-ching AI as a decision intelligence layer** that transforms fragmented and unstructured SME data into clear, actionable insights. Instead of relying on raw data or complex analytics, the system integrates **structured inputs** with **unstructured business context**, then preprocesses and filters this information to retain only decision-relevant signals. By combining **deterministic calculations (such as profit and opportunity cost)** with **AI-powered reasoning (Ilmu GLM-5.1)**, the system delivers explainable recommendations while maintaining consistency and accuracy. At the same time, strict input constraints, summarization, and data optimization ensure the solution remains **cost-efficient, fast, and scalable**, making it practical for real-world SME usage.

As a result, there is a clear gap for a system that can transform messy, real-world SME data into **clear, explainable, and actionable decisions**.

1.1. Purpose:

The system analysis document outlines the **technical scope, design decisions, and architectural approach** behind the development of **Ka-ching AI**. It defines how the system is structured to handle imperfect and unstructured SME data through a multi-level input framework, preprocessing pipeline, and decision intelligence layer. The document also explains key design choices such as limiting input size, filtering only decision-relevant data, and combining rule-based calculations with AI reasoning to ensure accuracy, consistency, and cost efficiency. Overall, it serves as a blueprint that justifies how Ka-ching AI is built to deliver scalable, reliable, and explainable business decision support.

So, the key element of system that has been covered are as followed:

1. **Architecture**:
   Ka-ching AI is built as a **lightweight, cloud-based modular system** consisting of a frontend (user input & results), backend (validation, preprocessing, and metric calculations), and an AI service layer powered by **Ilmu GLM-5.1**. The backend cleans and summarizes both structured and unstructured inputs, ensuring that only **decision-relevant data** is sent to the AI model. This modular design reduces cost, improves performance, and allows future scalability such as integration with Shopee, POS, or ERP systems.

2. **Data Flows:**
   User data flows from the frontend where inputs (options, cost, price, demand, notes, or CSV uploads) are collected, into the backend where it is validated, cleaned, summarized, and transformed into key metrics like profit and opportunity cost. Only the processed data is then sent to the AI model, which generates recommendations, risk analysis, and trade-offs. The results are returned to the frontend and displayed in a structured dashboard, ensuring fast, efficient, and accurate decision outputs.

3. **Model Process:**
   The model process begins with **input understanding**, where both structured data and unstructured notes are interpreted. The system then performs **data transformation**, converting inputs into standardized variables and calculated metrics. Next, **context integration** incorporates summarized business insights (e.g., risks, trends), followed by AI-driven **decision generation** that produces recommendations, explanations, and trade-offs. Finally, results are formatted into a **clear, user-friendly output**, ensuring decisions are explainable and actionable.

4. **Role of Reference:**
   This document serves as a high-level reference for developers and testers, outlining the system's architecture, data flow, and design decisions such as input constraints and preprocessing logic. It ensures all stakeholders have a shared understanding of how the system operates, enabling consistent development, testing, and future scalability.

1.2. **Background**:

Small and medium enterprises (SMEs) often struggle with making effective business decisions due to **fragmented data, limited analytical tools, and reliance on intuition rather than structured analysis**. Many business owners have access to raw information such as sales figures, product ideas, and market observations, but they lack a system that can transform this unstructured data into clear, actionable insights. Existing solutions are either too technical, requiring data expertise, or too generic, failing to reflect real business context and constraints.

To address this gap, **Ka-ching AI** was developed as a decision intelligence system that helps users evaluate business options in a structured and explainable way. It combines structured inputs (such as cost, price, and demand) with unstructured context (such as business notes and market conditions), and processes them through an AI-driven analysis layer. This allows SMEs to move from guesswork-based decisions to **data-informed, opportunity-aware, and explainable business decisions** in a simple and accessible manner.

1. Previous Version:
   In the previous approach, SMEs and online sellers typically relied on **manual and fragmented decision-making methods** such as spreadsheets, personal experience, and basic online tools. Data from sales, product ideas, and market observations were not centralized, making it difficult to analyze business performance holistically. Decisions were often made based on intuition rather than structured analysis, resulting in inconsistent outcomes, limited scalability, and frequent missed opportunities in pricing, product selection, and market timing.

2. Changes in Major Architectural Components
   In the Ka-ching AI system, the architecture has been transformed into a **centralized decision intelligence framework**. A new **AI-driven decision layer (Ilmu GLM-5.1)** has been introduced to process both structured inputs (cost, price, demand) and unstructured inputs (business notes, market context). A dedicated **preprocessing layer** is added to clean, validate, and summarize data before it reaches the AI model. Additionally, a **calculation layer** for metrics such as profit and opportunity cost is integrated, ensuring that decisions are not purely generative but also data-driven and consistent. This replaces the previous manual and disconnected workflow with a structured, automated pipeline.

3. New capabilities that are only introduced in this system
   Ka-ching AI introduces several new capabilities that were not available in traditional SME decision-making systems. It supports a **multi-level input system (Basic, Advanced, Pro)**, allowing users to choose between simple manual input and advanced dataset uploads. The system also introduces **unstructured input**

understanding, where natural language business notes are interpreted into structured decision signals. A key innovation is **opportunity cost analysis**, enabling users to compare multiple options and understand what they lose by not choosing alternatives. In addition, the system provides **explainable AI outputs**, where every recommendation is supported with reasoning, risk analysis, and trade-offs, making decisions transparent, trustworthy, and easy for SMEs to understand.

1.3. Target Stakeholder

Stakeholders	**Roles**	**Expectations**
SME Owners / Entrepreneurs	Primary users who input business options and make decisions	Want fast, accurate, and easy-to-understand recommendations to improve profit and reduce risk
E-commerce Sellers	Use the system to evaluate products, pricing, and demand	Expect data-driven insights to identify trending products and maximize sales performance
Business Analysts / Advanced Users	Upload datasets and analyze detailed business performance	Expect deeper insights, accurate calculations, and the ability to handle structured data .
Developers	Build, maintain, and improve the system	Need clear architecture, modular design, and scalability for future enhancements
Future Integration Partners (Shopee, POS, ERP providers)	Provide external data sources and system integration	Expect a stable, secure, and compatible system for API integration
QA Team	Test the system for bugs, validate outputs, and ensure reliability of AI-driven decisions	Expect stable system behavior, consistent outputs for similar inputs, clear test cases, and proper handling of edge cases

2. System Architecture & Design

2.1. High Level Architecture

2.1.1. Overview:

Type	Details
System	Web-based Decision Intelligence Platform
Architecture	Cloud-based modular architecture with separated layers

Ka-ching AI is structured as a **cloud-based client–server system** where users interact through a web-based application designed for SMEs. The system primarily consists of a single client interface. The user decision dashboard which communicates with the backend through an API layer. This interface allows users to input business options, provide relevant data such as cost, price, and demand, and optionally include unstructured business notes or upload datasets.

On the backend, the system is organized into **modular service components** to ensure flexibility and scalability. The **Input Processing Service** is responsible for validating, cleaning, and summarizing both structured and unstructured inputs. The **Decision Engine Service** performs deterministic calculations such as profit estimation and opportunity cost analysis, ensuring that outputs are grounded in quantitative logic. The **AI Integration Service** connects to Ilmu GLM-5.1, which generates recommendations, risk assessments, and trade-off explanations based on the processed data.

All services communicate through a centralized backend workflow, ensuring that only **decision-relevant and optimized data** is passed to the AI model. The system is deployed on a **cloud platform**, enabling accessibility, fast response times, and scalability. This architecture is intentionally designed to be lightweight and cost-efficient, while remaining extensible for future enhancements such as integration with e-commerce platforms, POS systems, and ERP solutions

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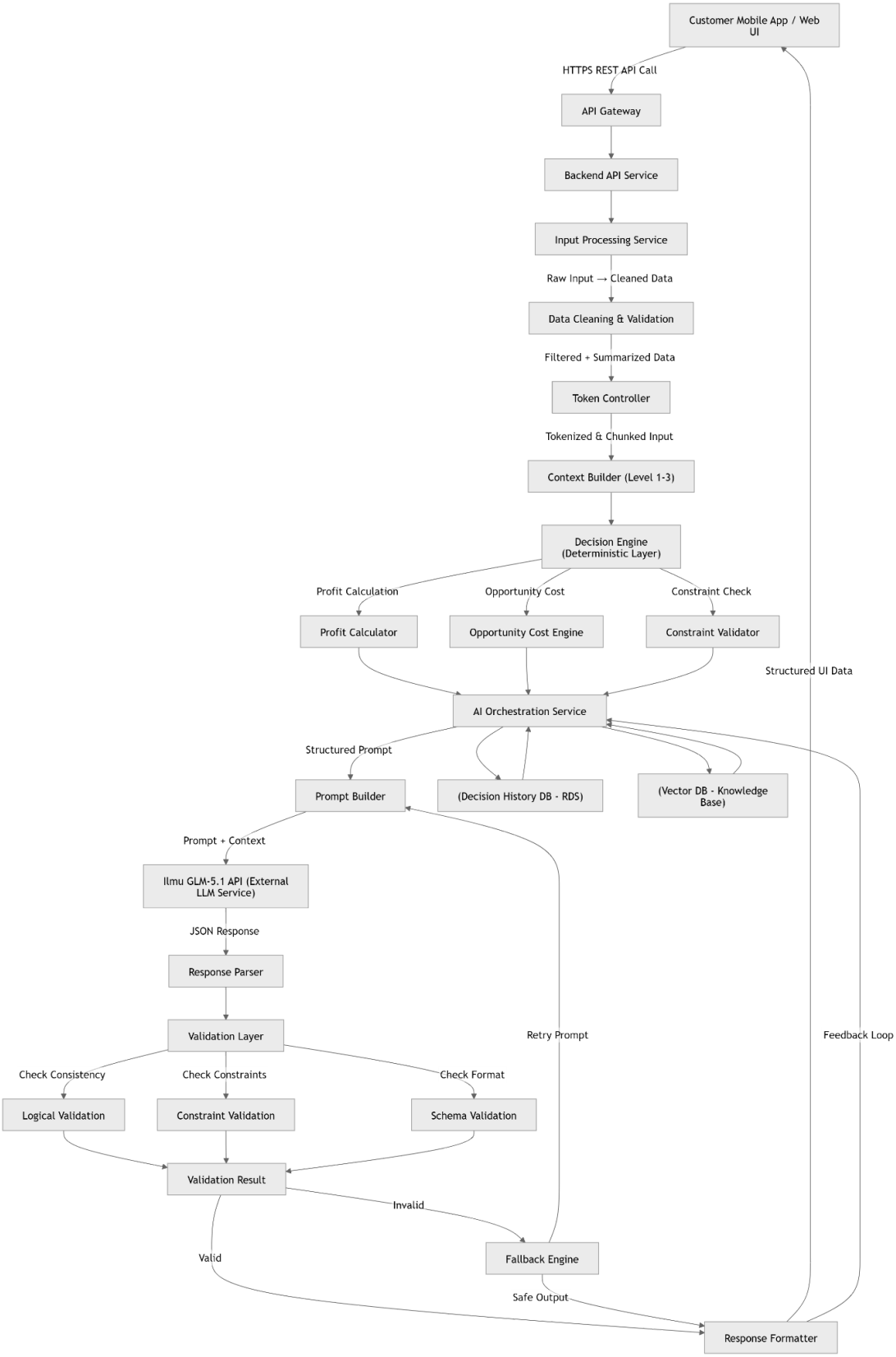
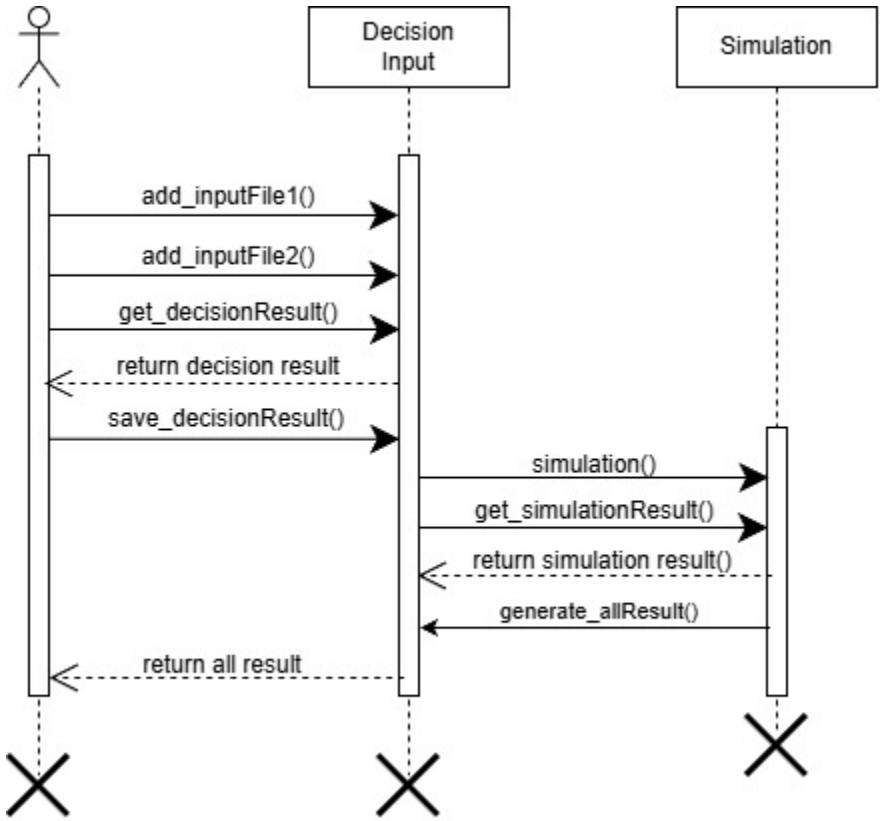
2.1.2 LLM as Service Layer

The architecture of your product needs to show how GLM is integrated as a service layer- not limited to a generic “AI Module”. The following Dependency diagram includes:

2.1.3 Dependency Diagram



2.2.2. Sequence Diagram



2.2. Technological Stack

2.2.1. Frontend

React.js – Web-based interface for user input (Basic, Advanced, Pro modes) and displaying structured decision outputs; lightweight and fast for rapid prototyping

2.2.2. Backend

Node.js (Express.js) – Handles RESTful APIs for input validation, preprocessing, summarization, and communication with the AI service layer

2.2.3. Database

No heavy persistent database in MVP (stateless design) – Only temporary/session-based data handling; future-ready for integration with PostgreSQL or MongoDB for storing user history and datasets

2.2.4. Cloud/Deployment

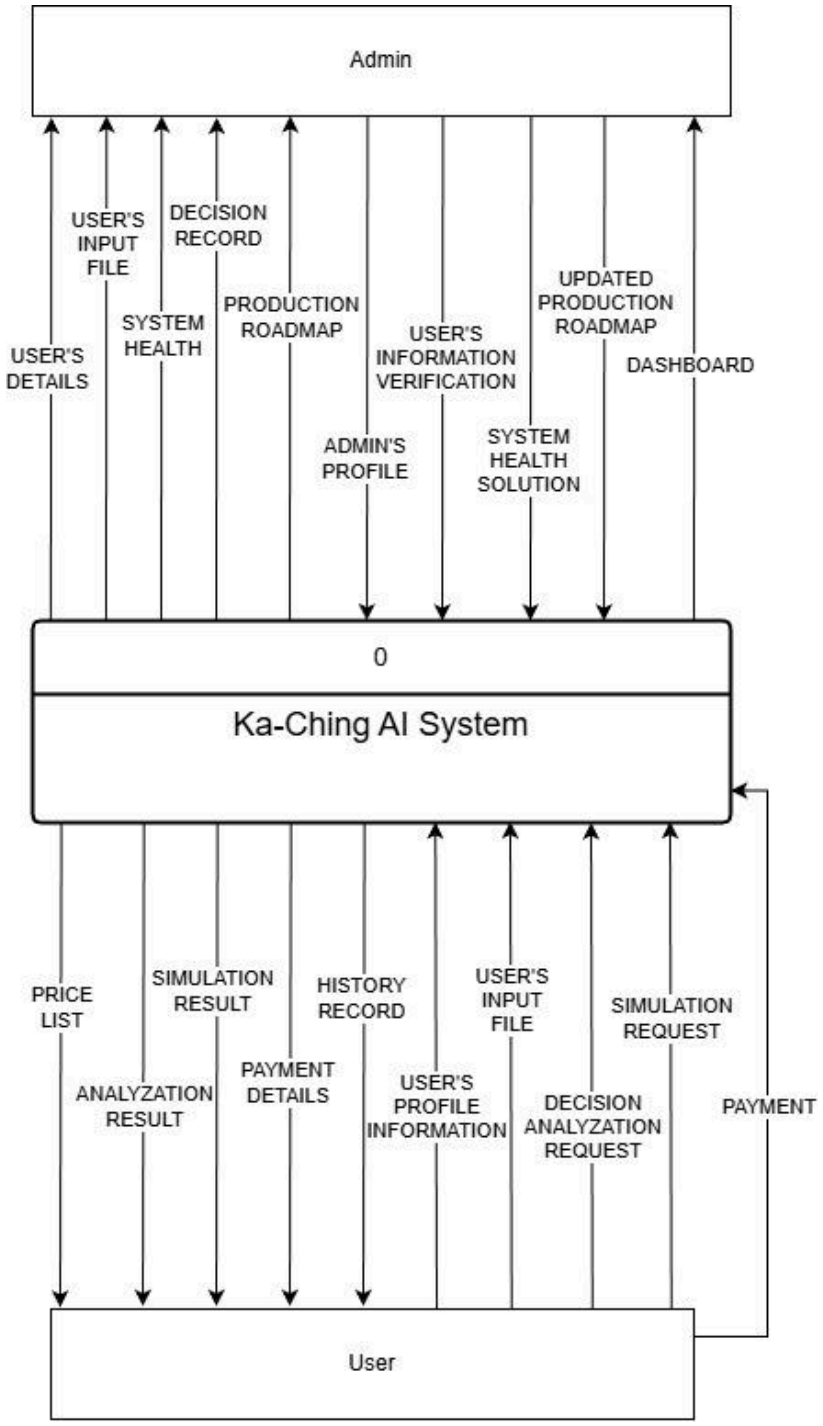
Vercel – Frontend hosting and deployment
Cloud backend services (e.g., serverless functions) – For scalable API handling

2.3. Key Data Flows

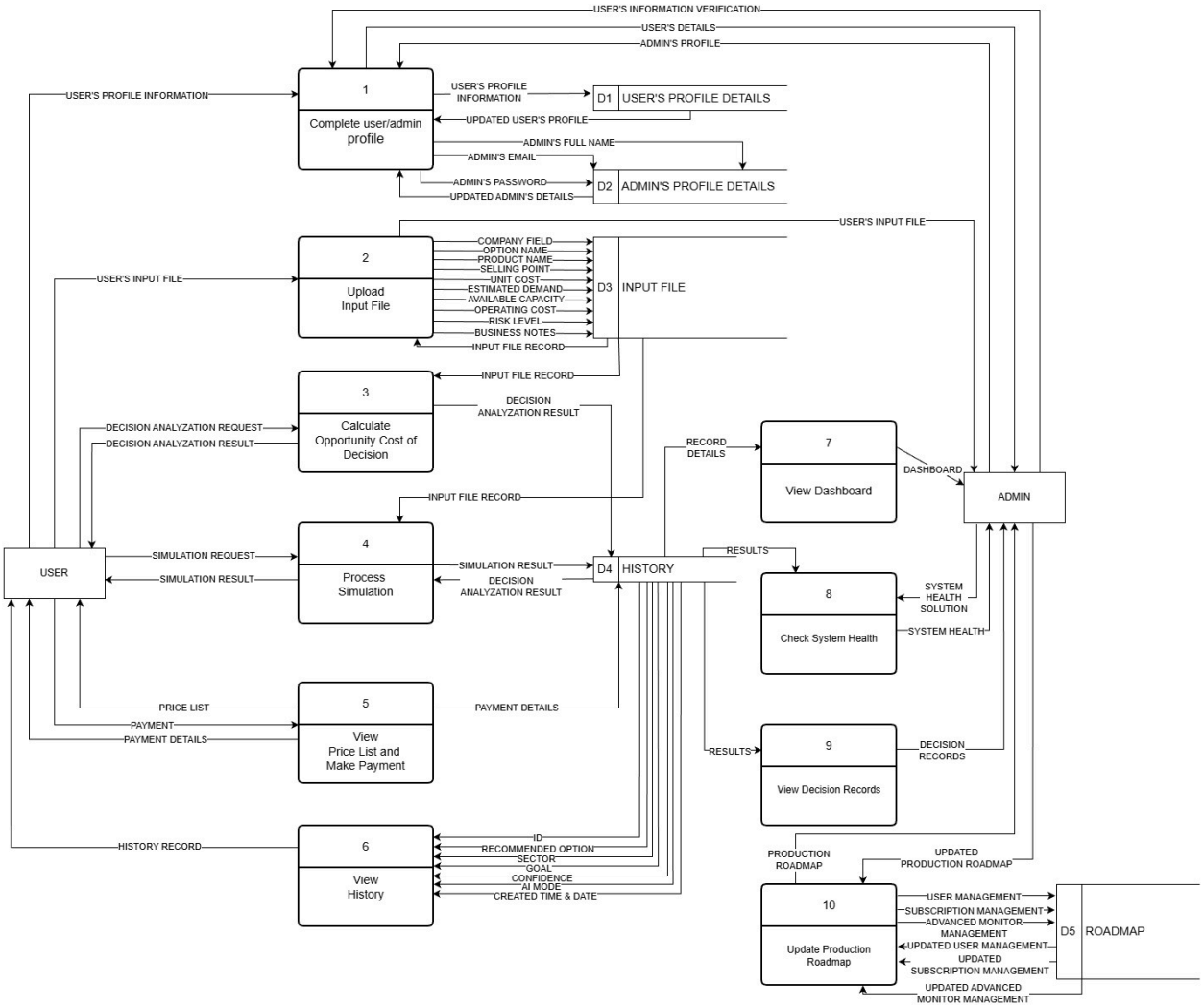
2.3.1.

2.3.1.1. Data Flow Diagram (DFD)

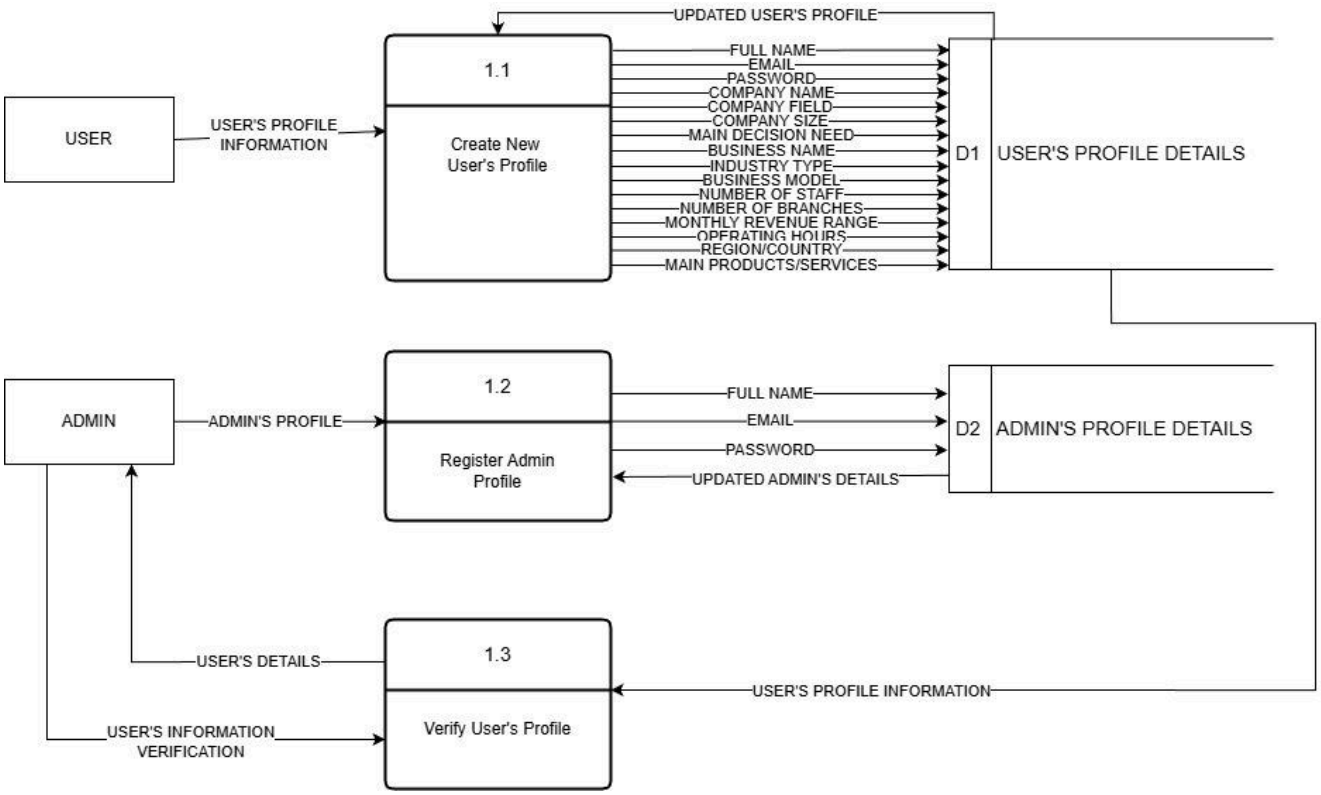
Context Diagram



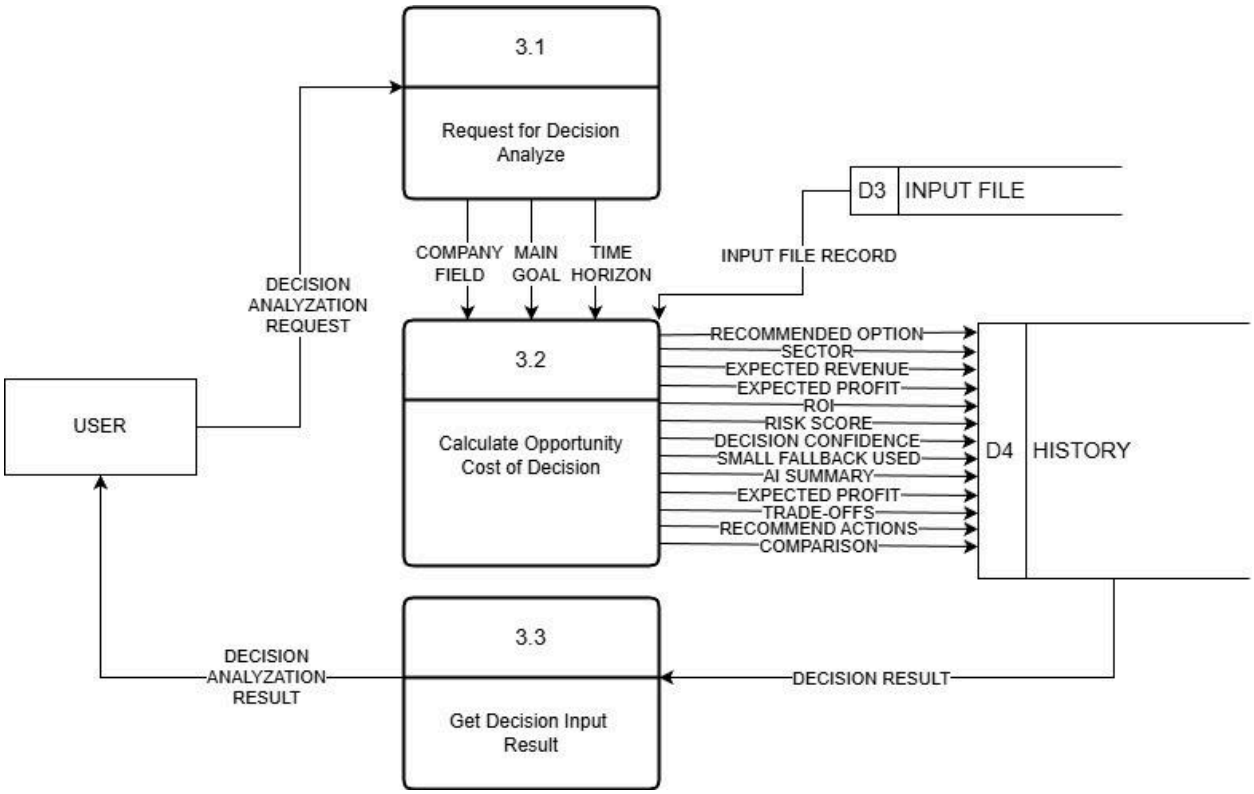
DFD Level 0


USER'S INFORMATION VERIFICATION
USER'S DETAILS
ADMIN'S PROFILE
USER'S PROFILE INFORMATION
1
Complete user/admin profile
USER'S PROFILE INFORMATION
D1 USER'S PROFILE DETAILS
UPDATED USER'S PROFILE
ADMIN'S FULL NAME
ADMIN'S EMAIL
ADMIN'S PASSWORD
UPDATED ADMIN'S DETAILS
D2 ADMIN'S PROFILE DETAILS
USER'S INPUT FILE
2
Upload Input File
USER'S INPUT FILE
COMPANY FIELD
OPTION NAME
PRODUCT NAME
SELLING POINT
UNIT COST
ESTIMATED DEMAND
AVAILABLE CAPACITY
OPERATING COST
RISK LEVEL
BUSINESS NOTES
INPUT FILE RECORD
D3 INPUT FILE
INPUT FILE RECORD
3
Calculate Opportunity Cost of Decision
DECISION ANALYZATION RESULT
DECISION ANALYZATION REQUEST
DECISION ANALYZATION RESULT
RECORD DETAILS
7
View Dashboard
DASHBOARD
ADMIN
INPUT FILE RECORD
USER
SIMULATION REQUEST
SIMULATION RESULT
4
Process Simulation
SIMULATION RESULT
DECISION ANALYZATION RESULT
D4 HISTORY
RESULTS
8
Check System Health
SYSTEM HEALTH SOLUTION
SYSTEM HEALTH
PRICE LIST
PAYMENT
PAYMENT DETAILS
5
View Price List and Make Payment
PAYMENT DETAILS
RESULTS
9
View Decision Records
DECISION RECORDS
HISTORY RECORD
6
View History
ID
RECOMMENDED OPTION
SECTOR
GOAL
CONFIDENCE
AI MODE
CREATED TIME & DATE
PRODUCTION ROADMAP
UPDATED PRODUCTION ROADMAP
10
Update Production Roadmap
USER MANAGEMENT
SUBSCRIPTION MANAGEMENT
ADVANCED MONITOR MANAGEMENT
UPDATED USER MANAGEMENT
UPDATED SUBSCRIPTION MANAGEMENT
UPDATED ADVANCED MONITOR MANAGEMENT
D5 ROADMAP

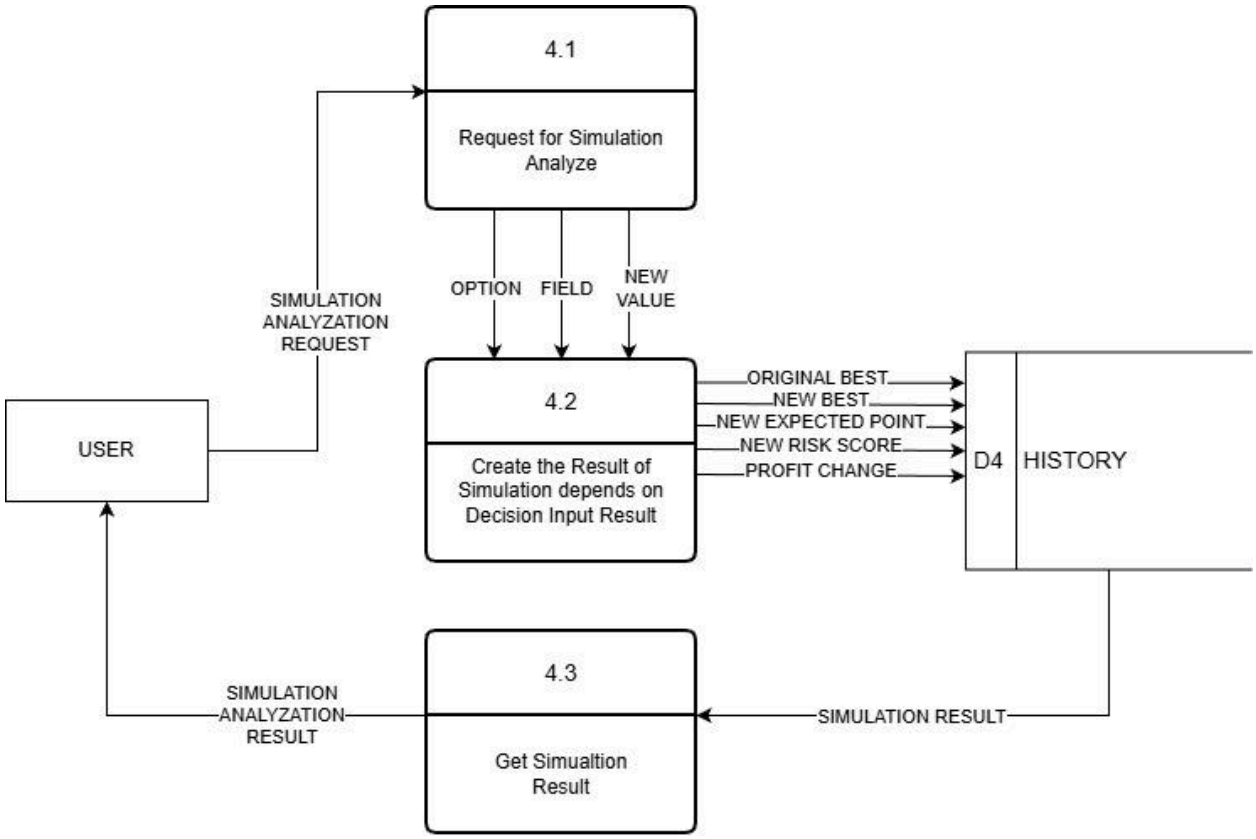
DFD Level 1 for P1



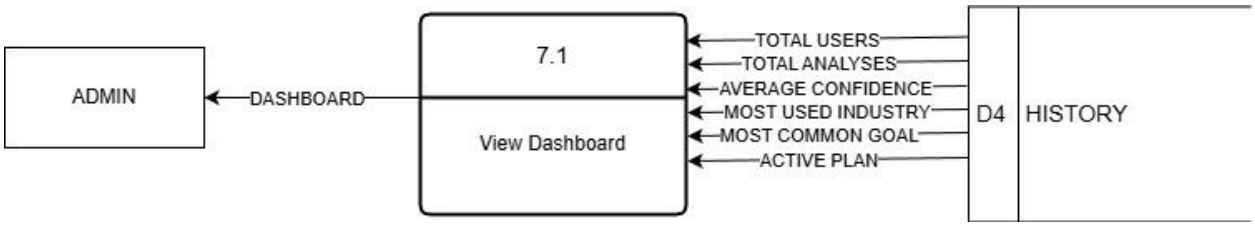
DFD Level 1 for P3



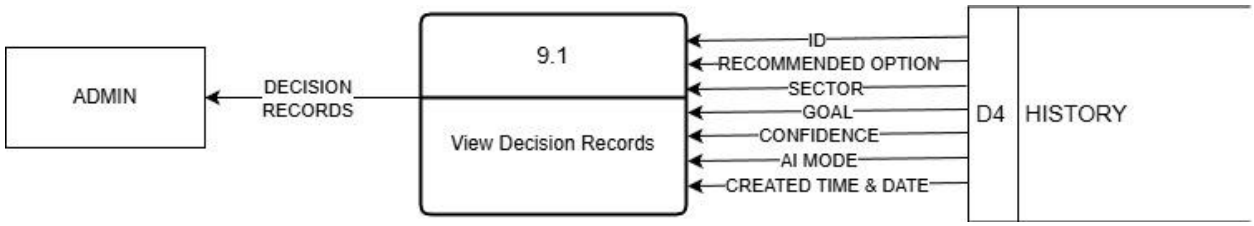
DFD Level 1 for P4



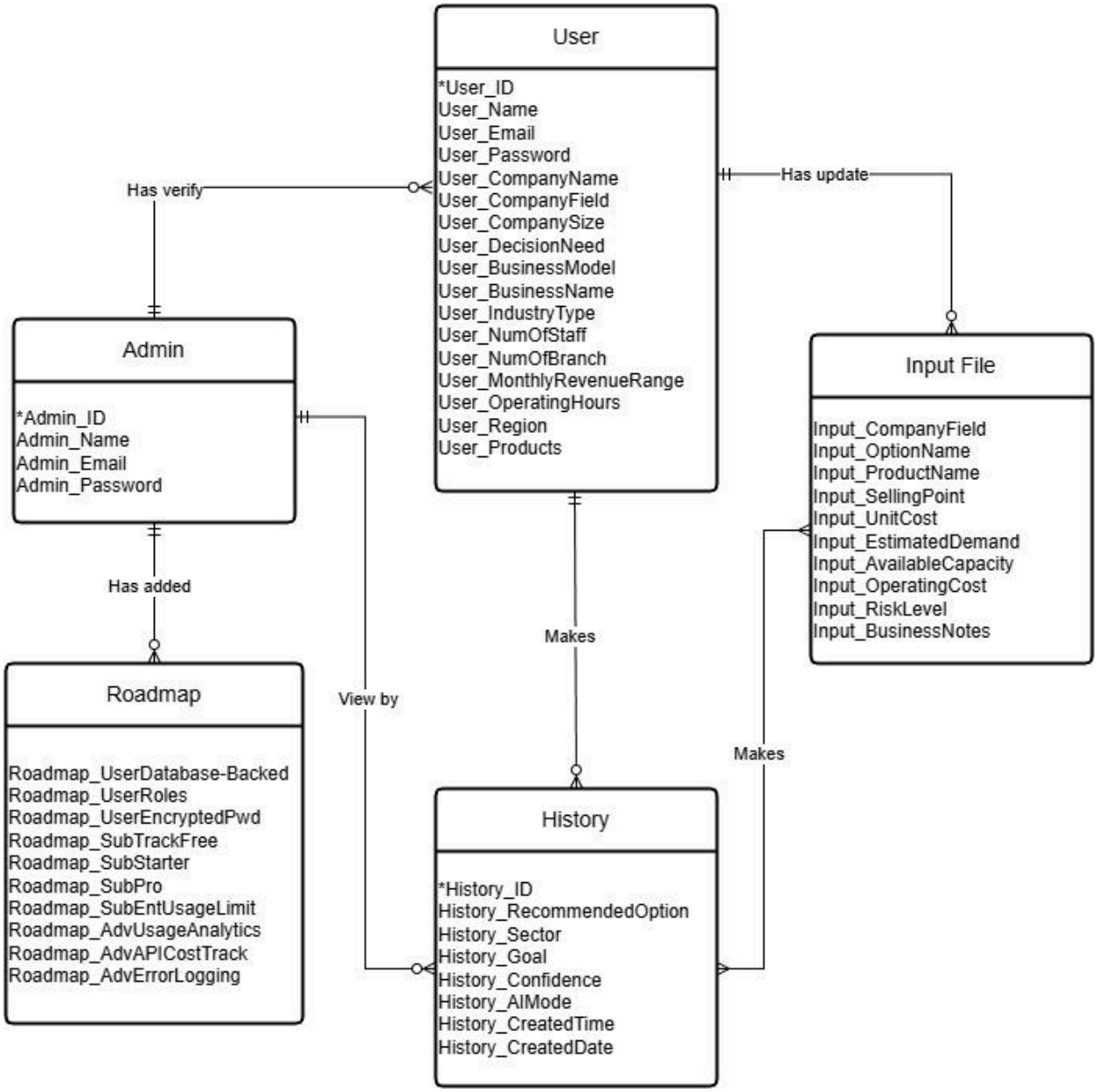
DFD Level 1 for P7



DFD Level 1 for P9



2.3.2. Normalized Database Schema



3. Functional Requirements & Scope

3.1. Minimum Viable Product:

3.1.1.

#	Features	Description
1	AI-Powered Project Cost & Resource Estimator	A decision-support tool that analyzes project requirements to calculate necessary raw material costs, labor hours, and machinery usage. It uses Z.AI GLM to perform Trade-off Analysis, such as determining if paying for overtime is more cost-effective than delaying a lower-priority order.
2	Interactive "What-if" Decision Lab (Scenario Simulation)	A simulation environment where Admins can test how external changes like a 15% increase in material prices or a machine breakdown impact project profitability and delivery timelines. The GLM explains the logic behind these shifts in a clear, "CEO-friendly" manner.
3	Dynamic Opportunity Cost Dashboard	A strategic module that ranks competing projects based on profit margins, ROI, and risk scores. It quantifies the Opportunity Cost of accepting one project over another, helping Admins optimize long-term revenue rather than just immediate tasks.
4	Decision Intelligence Engine	Core module that analyzes multiple business options to calculate profit, ROI, risk, and opportunity cost, helping SMEs select the best decision.
5	AI-Powered Reasoning (Ilmu GLM-5.1)	Generates structured explanations, trade-offs, and recommended actions based on both numerical inputs and business context.
6	Scenario Simulation ("What-if" Analysis)	Allows users to modify variables (e.g. cost, demand) and evaluate the impact on decision outcomes and profitability.
7	Real-World Market Intelligence Layer	Retrieves latest external market data to generate trend signals and strengthen decision recommendations with cited sources.

8	Results Dashboard & Visualization	Displays recommendations, comparison charts, decision confidence, and AI insights in a clear and user-friendly interface.

3.2. Non-Functional Requirements (NFRs)

Quality	**Requirements**	**Implementation**
Scalability	The system must handle simultaneous complex cost simulations from multiple SME Admins during peak planning periods.	Utilize cloud-based auto-scaling and load balancing to provision resources dynamically as session traffic increases.
Reliability	Project costings and resource plans must be 99.9% accurate and logically consistent with real-world business constraints.	Implement an automated validation layer to check GLM outputs against hard constraints like machine capacity and budget limits before display.
Data Privacy	All sensitive SME financial data and project costs must be encrypted and protected from unauthorized access.	Implement data sanitization and encryption at rest, ensuring only decision-relevant fields are processed by the GLM.
Performance	Complex project simulations and cost reports must return a response within 8 seconds to ensure a productive user experience.	Use asynchronous API processing and streaming text to provide immediate feedback while the full financial analysis is being generated.
Cost Efficiency	The inference cost per user session must be minimized to ensure the system remains affordable for small firms.	Implement caching for repeated market data queries and truncate oversized "messy data" inputs to save on token usage.

3.3. Out of Scope / Future Enhancements

A. Industry Expansion
   - Global Market and Policy Support
     Extending the engine to handle international trade regulations, multi-currency conversions and global tax compliance for SMEs involved in exporting and extending the global market for SMEs.
   - Multi-Industry Adaptation
     Expanding the decision logic to cater to non-manufacturing sectors like Retail, Healthcare with specific industry-tuned GLM modules.

B. Automation and System Integration (/)
   - Automated Procurement Execution
     A feature where the system not only recommends a budget but also automatically places order with suppliers when raw material levels hit a critical threshold.
   - Direct ERP/IoT Integration
     Real-time data syncing by connecting directly to factory floor IoT sensors and enterprise resource planning (ERP) systems to eliminate all manual data input.

C. Global Expansion & Compliance
   - Currency Conversion and Global Tax Compliance
     Support for multi-currency transactions and regulatory compliance to help SMEs expand into international markets
   - Multi-industry Adaption
     Extension of decision logic to support diverse sectors such as Retail and Healthcare through industry-specific GLM modules.

D. Advanced Intelligence & Collaboration
   - AI Supplier Negotiation Agent
     A chatbot capable of negotiating supplier pricing based on cost analysis and market conditions.
   - Predictive Maintenance Module
     Integration of machine sensor data to predict equipment failures and incorporate downtime risks into decision simulations.

E. Ecosystem & Platform Expansion

- B2B Marketplace Integration
  A platform for SMEs to trade excess production capacity or unused inventory to reduce waste
- Investor / Bank Dashboard
  A simplified financial and performance dashboard to support loan applications and investor evaluation.

4. Monitor, Evaluation, Assumptions & Dependencies

4.1. Technical Evaluation:

4.1.1. Grayscale Rollout & A/B Testing

We will adopt a phased release strategy. Initially, the "Cost Items and Strong Burst" feature will only be released to the top 100 small and medium-sized manufacturing enterprises, focusing on monitoring the match between AI-generated resource allocation suggestions and users' actual needs. We will conduct comparative tests between "pure data calculation" and "Z.AI GLM enhanced decision suggestions," aiming to assess the improvement in production scheduling efficiency for the fewest users adopting AI suggestions.

The system adopts a phased (grayscale) rollout strategy to ensure stability and reliability during deployment. New features, such as the Market Intelligence Layer and AI-enhanced decision engine, are initially released to a small group of SME users for controlled testing. During this stage, the system continuously monitors the accuracy of AI-generated recommendations, their alignment with user business needs, and overall system performance such as response latency. In addition, A/B testing is conducted by comparing pure data-driven calculations with Ilmu GLM-5.1 enhanced decision suggestions. This allows the team to evaluate improvements in decision efficiency, recommendation quality, and user adoption. Only after the system demonstrates stable and reliable performance will the feature be gradually rolled out to all users.

4.1.2. Emergency Rollback (ER) & Golden Release:

If a logical error occurs while processing "garbled data" or generating "what-if analysis" such as a negative material calculation, the system will trigger an automatic alarm and automatically stop all the prediction. Rollback Mechanism: The system will automatically roll back to a verified stable version (Golden Release), ensuring that administrators still have access to basic production data and historical records, unaffected by the faulty module.

To maintain system reliability, an emergency rollback mechanism is implemented to handle unexpected failures or anomalies. If errors occur during processing, such as invalid outputs, inconsistent calculations, or system instability, the system will automatically trigger an alert and suspend the affected processes. The rollback mechanism then restores the system to a verified stable version, known as the Golden Release. During this process, faulty components such as the AI reasoning or simulation modules are temporarily disabled, while core functionalities like basic calculations and access to historical data remain available. This approach ensures continuous system availability, prevents incorrect decision outputs, and maintains user trust even during system failures.

4.1.3. Priority Matrix:

Priority	Trigger Condition	Action
P1 Critical	Ilmu GLM-5.1 API failure or system crash	Activate fallback scoring engine and ensure system remains operational (graceful degradation)
P2 High	Market intelligence retrieval failure or no valid sources returned	Disable market layer and proceed with internal decision logic
P3 Medium	High latency or slow response (>5 seconds)	Reduce external calls and use cached results to maintain performance
P4 Low	Minor UI issues or non-critical display errors	Log issue and fix in next deployment cycle

4.2. **Monitoring**:

- **Real-time Tracking:** The system monitors Ilmu GLM-5.1 API response latency, success rate, and request usage to ensure stable performance.
- **Market Intelligence Monitoring:** Tracks success rate of external data retrieval (e.g. Google News RSS) and validates availability of relevant sources.
- **Health Check:** An automated health check runs periodically (e.g. every minute) to verify backend availability and core service functionality.
- Trigger & Response:
    - If API failure rate or latency exceeds threshold → system activates fallback scoring mode.
    - If market data retrieval fails → system proceeds without external data.
    - If the backend is unresponsive → alert is triggered for system recovery.

4.3. Assumptions:

4.3.1 SMEs have access to basic business data (e.g. cost, pricing, demand, or operational inputs) required for analysis.

4.3.2 Users provide reasonable and approximate inputs, even if not perfectly accurate.

4.3.3 Users have stable internet connectivity to interact with the system and the Ilmu GLM-5.1 inference engine.

4.3.4 SMEs have access to digital devices (laptop or smartphone) to use the web application.

4.3.5 Ilmu GLM-5.1 API is available and responsive during normal operation.

4.3.6 External market data sources (e.g. Google News RSS) reflect general trends, not guaranteed outcomes.

4.3.7 Users treat the system as a decision support tool, not a fully automated decision maker.

4.3.8 The system operates within controlled input limits (e.g. maximum number of options and realistic input ranges) to ensure performance and reliability.

4.4. **External Dependencies:**

Tools	**Purpose**	**Risks**
Z.AI Ilmu GLM-5.1 API	Serves as the core LLM reasoning engine that handles prompt inference, multi-step reasoning for resource allocation, and generates structured business insights, trade-offs, and recommendations.	High: In case of GLM API rate limitation or downtime, the core decision intelligence will fail. Mitigation: Implement a graceful degradation state using cached historical data and 3-level input strategies.
Google News RSS (Market Data Source)	Retrieves latest real-world market trends, product signals, and external business insights for decision enhancement	Medium – May return irrelevant or noisy data → mitigated with source ranking, filtering, and relevance scoring
Fetch API (Browser)	Handles communication between frontend (React) and backend (FastAPI)	Low – Standard web dependency with minimal risk
Vercel	Frontend deployment and hosting platform for React application	Low – Stable hosting, but build/deploy errors may affect UI availability
Render	Backend deployment and hosting for FastAPI services	Medium – Free tier may cause cold starts or temporary downtime
Local Storage (Browser)	Stores user role, profile, and session data for personalization	Low – Data is not persistent across devices and may be cleared by users

5. Project Management & Team Contributions

5.1. Project Timeline:

Timeline	Task	Justification
19 April - 20 April	Project Planning & Logic Design	Defining the idea with the problem that faced by SMEs especially the manufacturing, F&B and beauty product industry. Defining GLM prompt structures to ensure the reasoning engine is logically sound.
21 April - 23 April	Backend, GLM integration	Developing the multi-step agentic prompting flow and connecting the Z.AI GLM API for project cost estimation.
	Frontend & Dashboard	Building the “Scenario Simulation” UI and completing all the UI/UX design and system functions.
24 April - 25 April	Testing & Validation	Running "Constraint Validation" and "Logical Consistency" checks to ensure the AI recommendations are realistic.
	Demo Preparation & Final Pitch Video Preparation	Polishing the demo video and finalizing the documentation for submission.
26 April	Submission	Submit all the Document and Code Repository to the organizer.

5.2. Team Members Role:

Team Member	Role & Jobscope
Serene Lee Xuan Lin (Leader) Product Lead	Responsible for the Product Requirement Document (PRD), defining the core value proposition, stakeholder requirements, and the "What-if" decision logic for manufacturing SMEs.
Teoh Kai En System Architect	Responsible for the System Analysis Document (SAD), designing the multi-step agentic prompting workflows and the structural logic of the project cost estimation engine.
Loo Zheng Yi QA & Compliance	Responsible for the Quality Assurance Testing Document, ensuring the system adheres to non-functional requirements such as logical consistency, token latency, and budget constraint validation.
Ku Yee Ling Technical Lead	Responsible for the Code Repository (GitHub), implementing the Z.AI GLM API integration, backend microservices, and the interactive dashboard frontend.
Lim Chen En Pitcher	Responsible for the Preliminary Round Presentation Pitch Deck, translating complex technical features into a compelling business case focused on economic empowerment for SMEs.

5.3. Recommendations:

- Implement fallback mode to ensure the system remains functional when the Ilmu GLM-5.1 API is unavailable, maintaining reliability.
- Add data validation rules to prevent invalid or unrealistic business inputs that could affect decision accuracy.
- Introduce API response caching for market intelligence queries to improve performance and reduce repeated external requests.
- Apply source quality and relevance scoring to ensure only credible and relevant real-world data is used in decision-making.